still be covered within a few decades. What brain scientists are learning today about the amygdala and the cerebellum might make it possible for computers to outperform human psychiatrists and bodyguards in 2050.

AI not only stands poised to hack humans and outperform them in what were hitherto uniquely human skills. It also enjoys uniquely non-human abilities, which make the difference between an AI and a human worker one of kind rather than merely of degree. Two particularly important non-human abilities that AI possesses are connectivity and updateability.

Since humans are individuals, it is difficult to connect them to one another and to make sure that they are all up to date. In contrast, computers aren't individuals, and it is easy to integrate them into a single flexible network. Hence what we are facing is not the replacement of millions of individual human workers by millions of individual robots and computers. Rather, individual humans are likely to be replaced by an integrated network. When considering automation it is therefore wrong to compare the abilities of a single human driver to that of a single self-driving car, or of a single human doctor to that of a single AI doctor. Rather, we should compare the abilities of a collection of human individuals to the abilities of an integrated network.

For example, many drivers are unfamiliar with all the changing traffic regulations, and they often violate them. In addition, since every vehicle is an autonomous entity, when two vehicles approach the same junction at the same time, the drivers might miscommunicate their intentions and collide. Self-driving cars, in contrast, can all be connected to one another. When two such vehicles approach the same junction, they are not really two separate entities – they are part of a single algorithm. The chances that they might miscommunicate and collide are therefore far smaller. And if the Ministry of Transport decides to change some traffic regulation, all self-driving vehicles can be easily updated at exactly the same moment, and barring some bug in the program, they will all follow the new regulation to the letter.[4]

Similarly, if the World Health Organization identifies a new disease, or if a laboratory produces a new medicine, it is almost impossible to update all the human doctors in the world about these developments. In contrast, even if you have 10 billion AI doctors in the world – each monitoring the health of a single human being – you can still update all of them within a split second, and they can all communicate to each other their feedback on the new disease or medicine. These potential advantages of connectivity and updateability are so huge that at least in some lines of work it might make sense to replace *all* humans with computers, even if individually some humans still do a better job than the machines.

You might object that by switching from individual humans to a computer network we will lose the advantages of individuality. For example, if one human doctor makes a wrong judgement, he does not kill all the patients in